ENS3206/ENS6143 Power Systems 1

Design Project: Introduction to *PowerFactory*

Due: Friday the 17th of October 2025 at 23:59

Learning Outcomes

The aim of this project is to introduce the basic principles of the *PowerFactory* software. You will learn how to build a network model and enter the associated electrical data effectively. You will also be introduced to the Load Flow Calculation functions.

Instructions

- This Microsoft Word document is to be used as a template for the writing up the report of this design project. All requested answers should be typed, and high-resolution images placed if required, all in the indicated areas of this document. Reports with scanned hand-written content will **not** be accepted!
- Show all working. Answers to questions should be expressed clearly.
- Make sure the placeholder is replaced by your surname in the name of this Word document before submission.
- Submit your report via Canvas, by using the plagiarism-prevention service Turnitin. Extensions will not be granted without reasonable ground. Please refer to the ECU Policy on Assessment for details on late submission penalties. Do ***NOT*** use an assessment coversheet. A student declaration is now part of the submission process and coversheets impact Turnitin similarity reports.
- The total number of available marks is 20 (up to one mark per response).

Introduction

You can prepare for the project by getting familiar with the software before attempting the tasks below. *PowerFactory* is available in the computer labs in Building 5. You can use the *Getting Started* tutorials accessible via the *Help* menu.

In this project, you will learn how to use it for Short-Circuit Calculation. Fault calculations will be executed with different methods and the results will be analysed.

Task A. Creation of a First Project and Entry of Network Data

In this first exercise you will be introduced to the basic concepts of *PowerFactory*. A 110 kV grid will be built from scratch.

- To create a project, select the option *File* → *New*→ *Project...* from the main menu. Name the project with a unique name.
- A grid is automatically created in the new project and a dialog pops up. Name the grid *110 kV grid*. *PowerFactory* always needs at least one active grid.
- Also, a new diagram will be automatically created. Rename it Single Line Diagram HV. To rename the diagram, right click on the diagram tab and select → *Rename Diagram…*
- Change the drawing format from landscape to portrait using the icon *Drawing Format* from the graphic tool bar.
- You may maximise the graphic window by clicking on the icon *Maximise Graphic Window*. By clicking this icon again, you will get back to the original mode.

You are going to create the 110 kV grid shown in Figure 1. Here is some guidance about creating network elements in the single line diagram:

- Make sure that the *Graphical Freeze Mode* is not active. When a diagram is frozen no graphical changes can be made. This mode is usually used for evaluating results.
- Select an element from the Drawing Toolbox. You can configure the display of the *Drawing Tools* as you wish. The top of the *Drawing Tools* contains various display and filter options.
- Place the selected element in the network diagram by clicking with the left mouse button. If you want to enter the same element more than once, just click again in an empty space of the network diagram. If you want to select a different element, right click on the mouse, or press the **Esc** key.
- To move the element, click on it, keep the mouse button pressed and move the object toa new position. Releasing the mouse sets the new location.
- To resize a busbar or a terminal, click on the element to select it and drag one of the corners outwards.
- To connect an edge element (i.e., lines, transformers, loads, etc.), connect it directly to the required node when inserting it into the diagram. Transformers should always be drawn from the high voltage to the low voltage side.
- When editing the graphic, use the available graphic commands, in particular *Zoom In*, *Pan Tool*, *Zoom All*, and *Undo*.

- Double clicking on any element will open its dialog. Here you may input the different parameters.

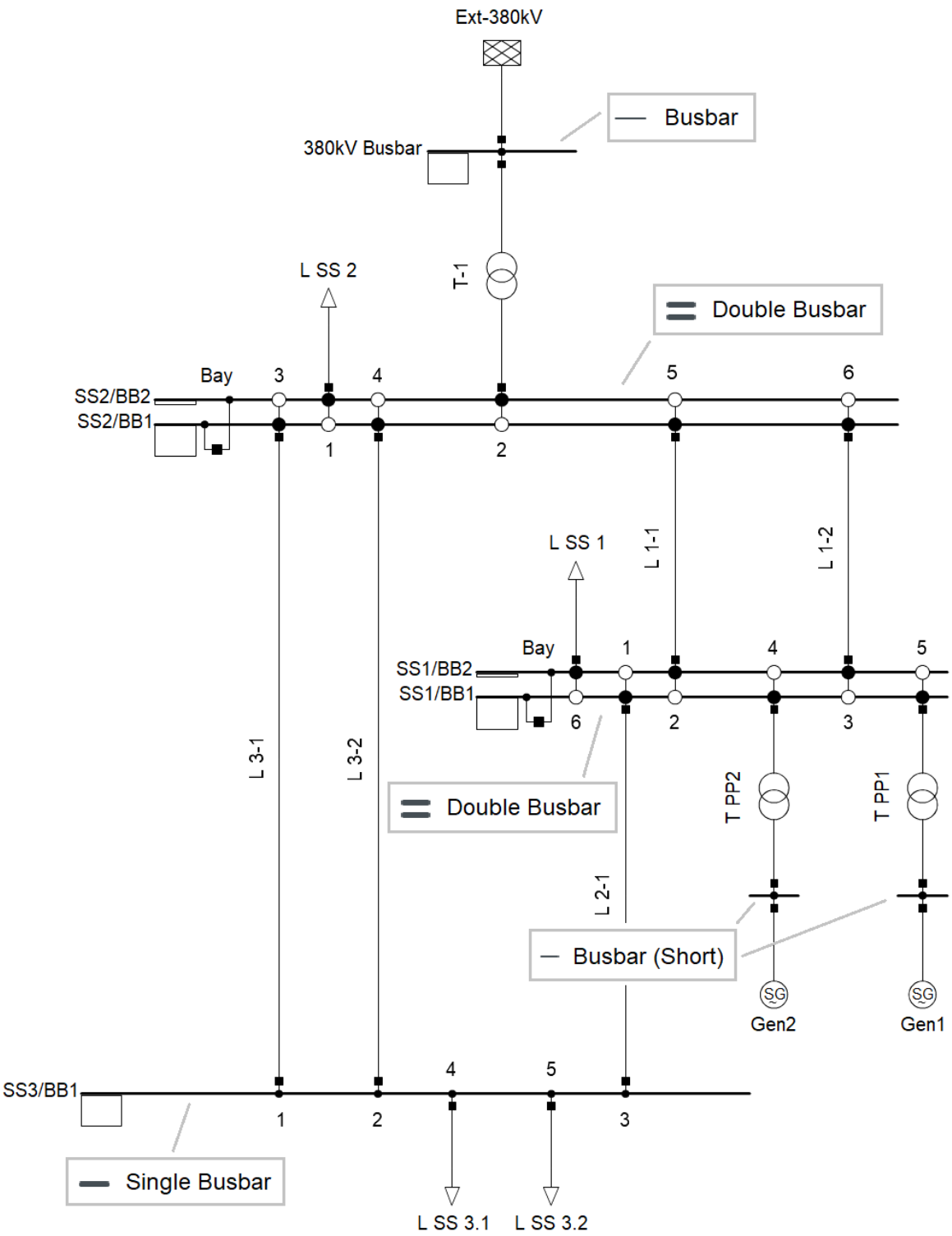


Figure 1: Single-line diagram of the modelled 110 kV network

In the following, the network model of a 110 kV transmission network is built using the graphic. The topology of the network is given in the Figure 1. Follow the instructions below.

- Always start with node positioning. Then it will be easier to connect Edge Elements. When editing the network, consider using the *Zoom* and *Undo* functionalities if necessary.
- Use the element *Busbar* to represent the 380 kV busbar. Change the voltage level accordingly.
- Use the *Busbar (Short)* element to represent the busbars between the generators and the transformers. The voltage level here is 10 kV.
- Use the elements *Single Busbar* and *Double Busbar* from the busbar systems to represent the 110 kV substations. Make sure the voltage level is set to 110 kV. When a busbar system is used, a substation is automatically created.
- Change the names of the single and double busbar objects according to Figure 1. To access the substation name, open the edit dialog of the busbar and then click on the arrow next to the *Substation* parameter.
- Connect the busbar systems with the *Line* elements and change the names according to Figure 1. Additional bays may be added by its detailed diagram can be opened by selecting the option *Diagrams → Show Detailed Diagram of Substation* from the context menu, resizing the busbars, copying existing bays (the numbered grey rectangles) and pasting them.
- Add the transformers, generators, loads and the external grid to the network and change the names accordingly. If you have incorrectly connected an item, right-click on the item and choose *Reconnect...* If an element is already connected correctly, but you want to redraw it, right-click on the element and choose *Graphic Object → Redraw…* By default, all power system elements are positioned "bottom down". If the element has already been placed and one wishes to flip it to the other side of the terminal, it can be done by selecting the element and then *right-click → Graphic Object → Flip at Node.*

So far, you have entered all the network elements, however some elements (e.g., lines, transformers, generators) are not completely defined without a type and need additional electrical information. Therefore, new types have to be defined and linked to the elements. You will create new types according to data sheets. The names of the individual objects assigned in the following steps are as shown in Figure 1.

- First you will create a new cable type. Select the line *L 2-1* with a double-click. You might have to change between the different pages of the element edit dialog to find the parameters. Basic parameters are specified on the *Basic Data* page. On this page, the input field for the *Type* is directly under the name. Press the down arrow button and select *New Project Type… → Line Type (TypLne)*. To define the line type, enter the parameters from Table 1 on the corresponding pages of the type edit dialog. The meaning of some parameters is still unknown to you, at this stage this is an exercise in exploring the software. For some parameters, such as the *AC-Resistance R'(90º C)* and *Earth-Fault Current*, there are various ways in which the data could be defined, therefore, use the button *Options* to choose between different input possibilities. Type parameters that are not listed retain their predefined value. The newly created type will be located in the project library. Once the cable type is defined, click on the **OK** button. Set the *Length of Line* to 25 km and click on the **OK** button.

Table 1: Datasheet of the used cable type

Name	N2XS(FL)2Y 1x300RM
Rated Voltage	110 kV
Rated Current	0.594 kA
AC-Resistance R`(20º C)	0.061 Ohm/km
AC-Resistance R0`	0.245 Ohm/km
Reactance X`	0.144 Ohm/km
Reactance X0`	0.578 Ohm/km
AC-Resistance R`(90º C)	0.078 Ohm/km
Capacitance C`	0.15 µF/km
Earth-Fault Current	9.002 A/km
Max. Operating Temperature	90º C
Max. End Temperature	240º C
Rated Short-Time (1s) Current	42.9kA

- Assign the new cable type to lines *L 1-1* and *L 1-2*. To assign the type to a line, click on the down arrow button on the *Type* field and choose *Select Type… → Line Type (TypLne)*. Set the *Length of Line* for the two lines to 10 km.

- Create for the 380/110 kV transformer *T-1* a *2-Winding Transformer Type* according to Table 2. Also set the *Tap Position* of the transformer to 14.

Table 2: Datasheet of the used 380/110 kV transformer type

Name	T-380/110-150MVA
Rated Power	150 MVA
Rated Voltage	380 kV / 110 kV
Vector Group	YNyn0
Short-Circuit Voltage uk	16.94%
Copper Losses	1030 kW
Short-Circuit Voltage uk0	14%
No Load Current	0.15%
No Load Losses	150 kW
Tap Changer 1	
Type	Ratio/Asym. Phase Shifter
at Side	HV
Additional Voltage per Tap	0.62%
Neutral Position	14
Minimum Position	1
Maximum Position	27
Tap dependent impedance	
Short-Circuit Voltage uk	
Minimum Tap	18.73%
Maximum Tap	15.91%
Copper Losses	
Minimum Tap	1020 kW
Maximum Tap	1107 kW
Short-Circuit Voltage uk0	
Minimum Tap	14%
Maximum Tap	14%

- Create another *2-Winding Transformer Type* for the 110/10-kV transformers according to Table 3. To assign the type to a transformer, click on the down arrow button on the *Type* field and choose *Select Type…* Set the *Tap Position* for these transformers to 0.

Table 3: Datasheet of the used 110/10 kV transformer type

Name	T-110/10-60MVA
Rated Power	60 MVA
Rated Voltage	110 kV / 10 kV
Vector Group	Dyn5
Short-Circuit Voltage uk	12.5%
Copper Losses	144 kW
Short-Circuit Voltage uk0	10.5%
No Load Current	0.08%
No Load Losses	31.51 kW

- Define one *Busbar Type* for all 110 kV busbars in the network according to Table 4.

Table 4: Datasheet of the used busbar type

Name	SS 110 kV
Rated Voltage	110 kV
Rated Current	1.02 kA
Rated Short-Time Thermal Current	16 kA 1s
Short-Circuit Peak Current	30 kA

- Create a synchronous machine type for the generators according to Table 5. Set the *Local Controller* to *Const. V*, the *Active Power* to 30 MW, and the *Voltage* to 1 p.u.

Table 5: Datasheet of the used synchronous machine type

Name	46 MVA Gen
Rated apparent power	46.55 MVA
Rated voltage	10.5 kV
Rated power factor	0.8
Connection	YN
Subtransient reactance saturated value xd``sat	0.15 p.u.
Stator resistance rstr	0.0132 p.u.
Reactance x2	0.29 p.u.
Subtransient reactance xd``	0.175 p.u.
Transient reactance xd`	0.27 p.u.
Rotor type	Salient pole

- The external network has no type. Only element parameters shown in Table 6 must be entered.

Table 6: Relevant information on the modelled external grid

Name	*Bus Type*	*Voltage Setpoint*	*Angle*	*Short-Circuit Power Sk``max*	*Short-Circuit Power Sk``min*
Ext-380kV	SL	1.02 p.u.	0	8000 MVA	7000 MVA

- The element parameters of the loads are listed in Table 7.

Table 7: Relevant information on the modelled loads

Name	*Input Mode*	*Active Power*	*Power Factor*
L SS 1	P, cos(phi)	22 MW	0.86 ind.
L SS 2	P, cos(phi)	35 MW	0.86 ind.
L SS 3.1	P, cos(phi)	46.4 MW	0.85 ind.
L SS 3.2	P, cos(phi)	50 MW	0.85 ind.

There is also the possibility to assign types from the DIgSILENT Library to elements in the network. All available types of the global DIgSILENT Library are free to use. However, the types are read-only and can therefore only be used but not changed. If you want to edit types from the global library, first copy them into your project library.

- The two lines *L 3-1* and *L 3-2* shall be modelled with a pre-defined tower type from the global library. First, mark both lines by keeping Ctrl pressed. Right-click the selection and choose the option *Network Models → Line Couplings → New…* The *Data Manager* will open. Navigate through the data tree on the left pane until you reach the *DIgSILENT Library* and show the contents of the folder *Equipment Types → Towers*. Select the tower type *110 kV x2 -Donaumast- 2x Al/St 435/55* and press **OK**. A new dialog will open, where the lines are assigned to the circuits. Select both lines and click **OK**. Once the line coupling is defined, click on the **OK** button. Set the *Length of Line* for the two lines to 50 km.

- Use the command *Network Data Assessment* to check if all the data was correctly inserted and see if there are any problems with the network topology. Press the *Network Data Assessment* icon to open the edit dialog of the command. Select *Verification* and *Check → Input parameters*. *Execute* the command and check for messages in the output window. Execute the command again with the options *Verification* and *Check → Topology*. Fix the errors if there are any.

- There are a few additional options for further graphical customisation of your project. Make the title block visible by activating the *Title* layer within the *Layers* icon. Double-click the title block in the bottom right corner to edit it. Press the right-pointing arrow button to edit the *Title Text*. To display a company logo, you can select a file as *Graphic*. Also, add the *Vector groups* layer to the grid by selecting it under the *Element Details* of the *Layers* configuration.

- Export the single-line diagram to your pen drive as a *.pdf* file via the button *Export Diagram*.
- Via *File → Export→ Data (*.pfd;*.dz)…* , choose your project and export it to your pen drive.

Task B. Load Flow Analysis

- Import and activate your project. To import a project, select from the main menu *File → Import→ Data (*.pfd;*.dz;*.dzs)…* and then select the file to import. To activate it, select *Activate project after import*.
- From the main icon bar, choose the *Calculate Load Flow* command. Leave all settings as they are and *Execute* the Load Flow Calculation command. Check the messages in the output window to verify if there are any warnings or error messages (error messages will be shown in red colour) and correct the mistakes if there are any listed. Once mistakes are corrected, repeat the load flow calculation.
- If no error messages or warnings are displayed in the output window, the basic setup of the network model is fine. Show a screenshot of the single-line diagram of the modelled grid, with results of the performed load flow analysis, in Figure 2.

Place your answer here →

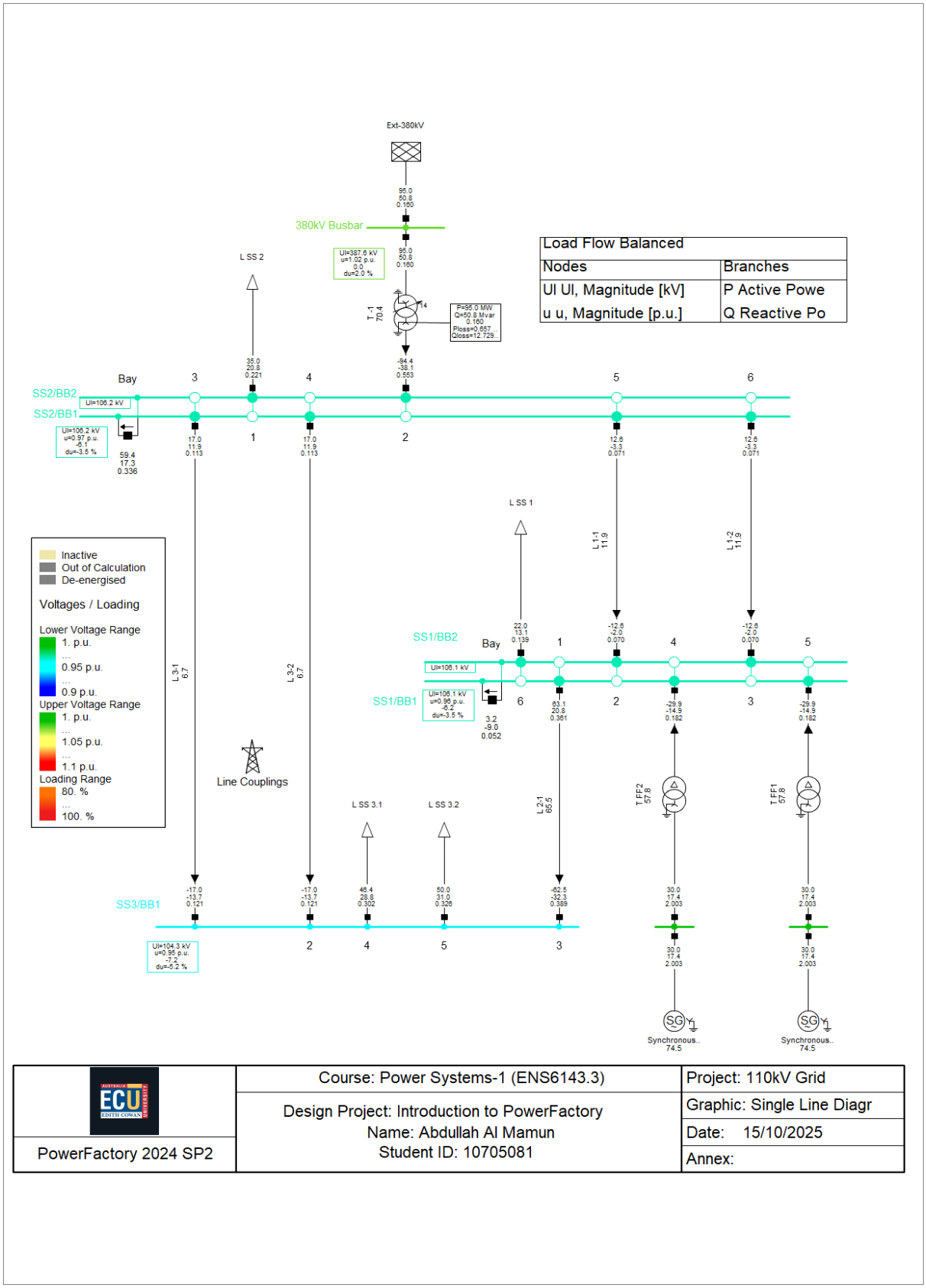


Figure 2: Single-line diagram of the modelled 110 kV network with results of the performed load flow analysis.

- Select the button *Colouring* to open the colouring dialog for colouring the network diagram. By using the button *Colouring* the colouring of the network graphic can be configured. The graphic colouring scheme for each calculation can be defined on the corresponding page (left). If no calculation results are active, the colouring scheme of the page Basic Data is used. Notice that after a load flow calculation, the check box *3. Other* is enabled automatically, with *Results* for the first selection and *Voltage Levels* for the second selection.
- Press the *Show Legend Layer* button to show the colour and results box legend. The result box legend usually appears in the bottom-left corner of the network diagram. You can move and resize the legend by selecting it.
- Use the *Layers* button and jump to *Element Details*, where the setting *Power flow direction arrows* has to be set. Configure *Direction represents* to show the direction of the *Active Power* flow.
- Right-click on a result box of a load and hover the mouse over *Format for Edge Elements*. Switch to *Current, Power Factor* to see the changes in the single line diagram (result boxes and legend). Set it back to *Branch Flow*. Select the *Branch Flow* result box format again and observe the changes in the listed variables.
- Change the exponent for the current from kA to A. The exponents of the units can be configured via the *Formats and Units* page in the dialog to the project settings. To open the dialog, go to the main menu and select *Edit* → *Project Data*→ *Project Settings...*
- Right-click on the result box of a busbar and select → *Edit Format for Nodes*. Right-click below the table and select *Append Row(s)*. Double-click on the field in the column corresponding to *Variable*. Select the parameter Voltage Deviation *m:du* from the *Variable Set Currents, Voltages, and Powers*.
- Right-click on the transformer *T-1* and select *Text Boxes* → *Create Additional Result Box*. This will create a default textbox in the single line diagram. Right-click on the new result box and select → *Edit Format*. Add the variables *c:Ploss* and *c:Qloss* from the *Variable Set Calculation Parameters*.
- Complete Table 8.

Place your answer here →

Table 8: Load flow analysis results

Lowest voltage in p.u. & Busbar name	0.95 p.u. SS3/BB1
Highest loading of a line & Line name	65.5% L-2-1
Active power on HV side of T1	95 MW
Reactive power on HV side of T1	50.8 MVar

- For a clear presentation of the results, the Network Model Manager can also be used. Here all calculation-relevant elements of the network are listed in classes. Click the *Network Model Manager* button in the main icon bar and display the *Busbar* calculations. Go to the *Flexible Data* tab at the bottom and sort the list according to the voltages in p.u. To sort the list, click on the corresponding column title. Filter the data to get only the busbars with a voltage below 1.00 p.u. Use the down arrow select icon to define the column filter. Show only two decimal places. To do so right-click on the column title and select *Edit Number Format...* The number of decimal places can also be configured globally - but project-specific - via the project settings. These are accessible on the *Flexible Data* page of the *Network Model Manager* via the *Formats and Units* button.
- Fill out Table 9 with the results. Report only one busbar per substation.

Place your answer here →

Table 9: Busbar voltages

Name	In Folder	Grid	Nom.L-L Volt. kV	UI, Magnitude kV	u, Magnitude p.u.
BB1	SS1	110 kV	110 kV	106.1 kV	0.96 p.u.
BB1	SS2	110 kV	110 kV	106.2 kV	0.97 p.u.
BB1	SS3	110 kV	110 kV	104.3 kV	0.95 p.u.

- Open the *Network Model Manager* and display *Grid*. On the *Flexible Data* page, click the *Variable selection* button and select the following variables to display:
    - *Generators, Active Power (c:GenP)*
    - *Loads, Active Power (c:LoadP)*
    - *Losses, Active Power (c:LossP)*
    - *Line Losses, Active Power (c:LossPlne)*

There are different *Variable Sets* for the results. For example, for lines, variables with the letter *m:* represent results that are calculated directly from the load flow (currents, voltages, and powers). The letter *c:* stands for the variable type *Calculation Parameter* and contains, for example, the loading for lines and loss quantities for network elements.

- Fill out Table 10 with the results.

Place your answer here →

Table 10: Results (Flexible Data) of the Grid

Name	Generators, P [MW]	Loads, P [MW]	Losses, P [MW]	Line Losses, P [MW]
110 kV	60.0 MW	153.4 MW	1.6 MW	0.8 MW

- By pressing the icon *Copy (with column headers)* you can copy marked table areas and insert them for example in Excel for further documentation.
- Are there elements with a loading higher than 60%?

Place your answer here →

There are four components which are T1 (70.4%), L2-1 (65.5%), Synchronous Gen-1 (74.5%), Synchronous Gen-2 (74.5%)

- Various predefined reports are available for evaluation and documentation of results. Click on the *Clear All* button in the *Output Window* to clear it. ASCII reports can be created by pressing the *Generate reports* button in the main menu. Create an ASCII report for the *Total System Summary* and observe the results in the Output Window. By pressing the button *Save Output File* you can save the content of the Output Window in a text file. With the icon *Print* you can save the Output Window as a pdf file.
- To check the Voltage Profiles in the network, the ASCII report *Voltage and Loading Violations* can also be used. Are there elements with voltage deviation exceeding ± 0.05 p.u.?

Place your answer here →

15/10/2025

Voltage and Loading Violations

Lower Voltage Violations

Lower Limit of Allowed Voltage 0.95 p.u.				
Name	**Element type**	**Station/Grid**	**Voltage, Magnitude p.u.**	**Line-Line Voltage, Magnitude kV**
BB1	Busbar	SS3	0.948	104.3

Figure: Voltage & Loading Violations

Task C. Voltage Control

As you may have realised, the voltage at some busbars in the 110 kV grid is relatively low. To support the voltage in the grid, you will firstly use a station control. Currently the generators are controlling the voltage at their local busbars. Now a station control shall be implemented to control the voltage in the network. A station control is a reactive power controller for one or more generators which will keep the voltage at a specific busbar at the given set-point.

- Define a station controller for the generators *Gen1* and *Gen2*. The station control should control the voltage of busbar *BB1* of the substation *SS1*. To define a new station controller, mark the two generators and the busbar. Right-click on the selection and select *Network Models* → *Station Control* → *New…* This will open a predefined *Station Control* dialog, which already has the generators entered at the corresponding positions. On the *Load Flow* page, set the *Voltage Setpoint* to 1.02 p.u. Under the *Distribution* tab, change the *Reactive Power Distribution* to be shared equally *According to Rated Power*. Use the *Info* button to print the control strategy in the output window.
- Execute the load flow calculation. Has the new voltage setpoint of *SS1* been reached?

Place your answer here →

The new voltage setpoint of SS1 has been reached and it was 1.02 p.u.

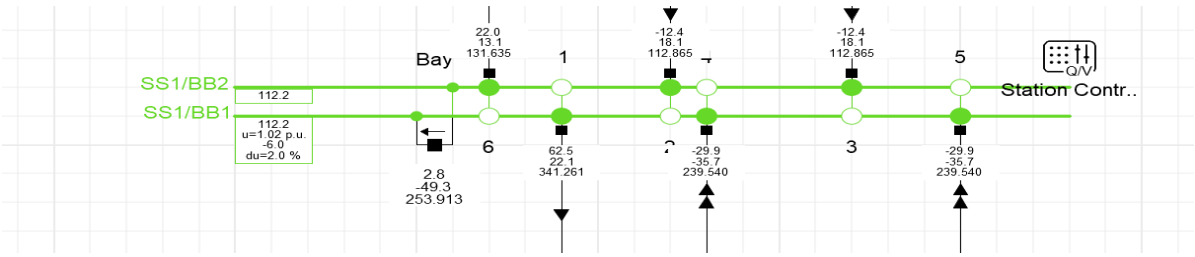


Figure: New Voltage Point of SS1

- Are there any overloaded elements?

Place your answer here → Yes, Both the Synchronous Gen- and Synchronous Gen-2 are 107.5%.

- Now we want to analyse the effect of the station controller on the network. Put the station controller *Out of Service*. The station controller is linked as an *External Station Controller* on the *Load Flow* page of the generators. You can also use the *Network Model Manager*. Execute a load flow calculation.
- We will now use the *Comparison of results* tool to compare the results with and without station control. Click on the *Comparing of Results on/off* button. A message indicating that the results were saved as Case A will pop up. Put the station control in service and execute the load flow calculation command again. The results are now deviations in percent relative to Case A. Click on the button *Edit Comparing of Results* and change the *Compare Mode* to *Case_B-Case_A*. What is the difference in the loading of the generators?

Place your answer here →

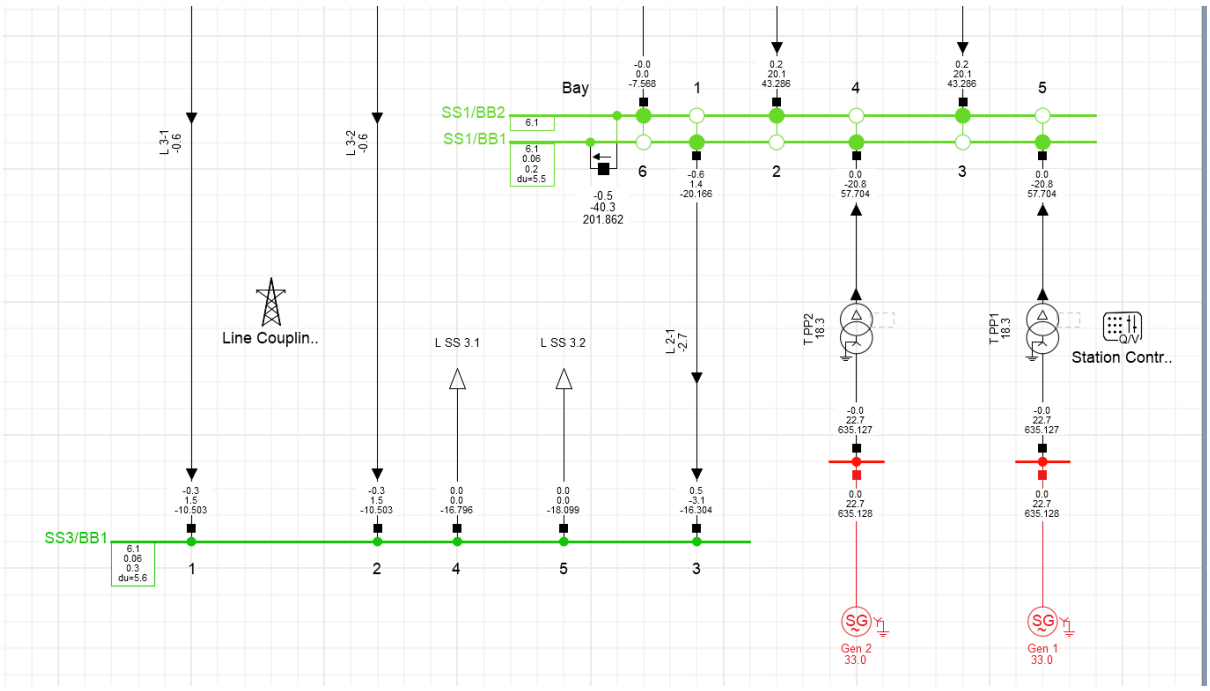


Figure: Loading of the Generators.

- Press again the *Comparing of Results on/off* button to disable the function.
- To prevent overloading of the generators, operating limits can be specified. Set the reactive power limits of the generators *Gen1* and *Gen2* to -12 Mvar and +30Mvar. The reactive power limits are fixed on the *Operational Limits* tab on page *Load Flow* of the synchronous machine dialogue. Execute a load flow calculation. Are there any warnings listed in the output window?

Place your answer here → Yes, they are given below.

```
Report of Control Condition for Relevant Controllers
-----------------------------------------------------------------------------
110 kV grid\Station Control.ElmStactrl:
 ' Gen 1' : Maximum reactive power limit exceeded ( Q = 40.08 Mvar > Qmax = 30.00 Mvar )
 ' Gen 2' : Maximum reactive power limit exceeded ( Q = 40.08 Mvar > Qmax = 30.00 Mvar )
-----------------------------------------------------------------------------
```

Figure: Warning

- Open the page *Load Flow* → *Operational Limits* for each generator to view its current operating point of the generator, as determined by the load flow calculation. Comment on its location relative to the capability curve.

Place your answer here →

The location of the operating point of each generator are located out of the capability curve.

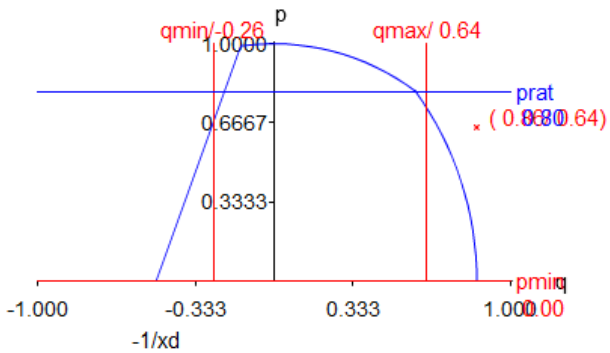


Figure: Capability Curve

- Click again *Calculate Load Flow* to open the *Load Flow Calculation* command and enable the option *Consider reactive power limits*. *Execute* the load flow. Are there any changes in the results? Can the voltage setpoint of the station control be reached? If not, why not?

Place your answer here →

Yes, the voltage setpoint decreased to 1.00 p.u. from 1.02 p.u. So, we can see that the voltage setpoint of the station control cannot be reached. Because at this point the operating point of generators are located out of the capability curve.

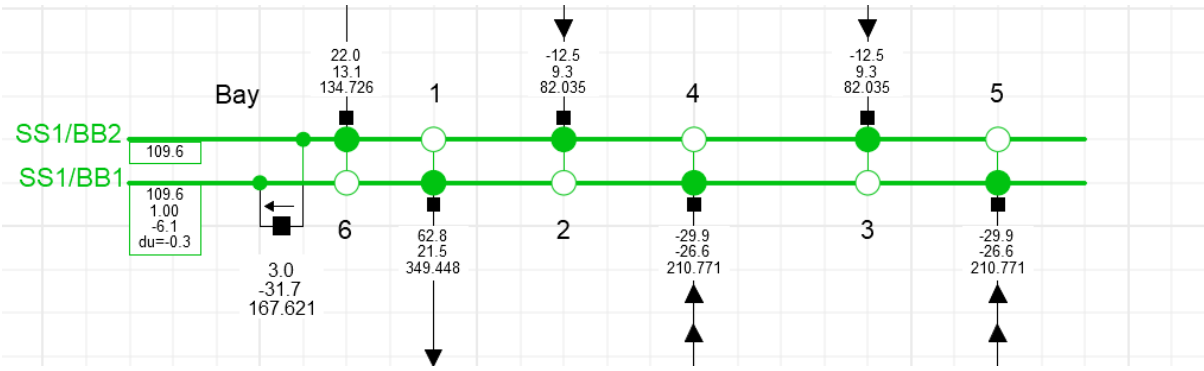


Figure: Voltage setpoint

- Reactive power limits of generators are actually dependent on the operating point. By considering the generator capability curve the reactive power limits can be entered precisely in PowerFactory. Create a new capability curve, which is dependent on the active power of the machine. Use the values given in Table 11. Open one of the generators and navigate to the page *Load Flow* → *Operational Limits*. *Select* via the down arrow button a new *Capability Curve*. The folder *Mvar Limit Curves* will automatically open. Use the *New Object* icon to create a new *Capability Curve*. On the page *Configuration* you can set the *Input Model* (MW/Mvar or p.u.) that shall be used for the definition. On the *Basic Data* page, insert the desired number of rows by right-clicking in the empty table and selecting *Append n Rows* to enter the values. Define the capability curve according to the table.

Table 11: Capability Curve - Active power with corresponding reactive power limits

Active Power p.u.	Min. React. Pow. p.u.	Max. React. Pow. p.u.
0	-0.5	0.86
0.3	-0.4	0.83
0.6	-0.29	0.72
0.8	-0.22	0.6
0.9	-0.18	0.4
0.95	-0.16	0.3
1	-0.1	-0.04

- Apply the new capability curve to generators *Gen1* and *Gen2*. Execute the load flow calculation again. Are there any warnings listed in the output window?

Place your answer here →

There is a warning which is given below.

```
     Report of Control Condition for Relevant Controllers
-------------------------------------------------------------------------------
110 kV grid\Station Control.ElmStactrl:
' — BB1' : Voltage setpoint cannot be kept ( v = 0.9125 p.u., vset = 1.0200 p.u. )
-------------------------------------------------------------------------------
```

Figure: Warning

- Open the page *Load Flow* → *Operational Limits* for generator *Gen1* to view the current operating point of the generator, as determined by the load flow calculation. What is its location relative to the capability curve?

Place your answer here →

Operating point is on the capability curve.

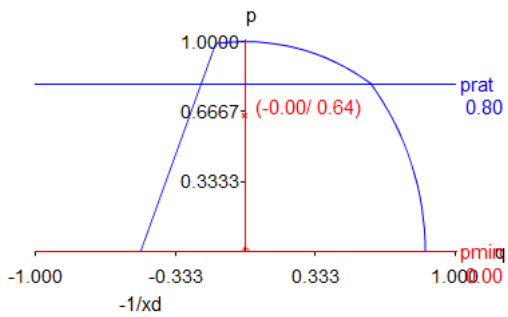


Figure: Operating point of Capability Curve

- The voltage can be additionally controlled by changing the tap position of a transformer. Verify that the Station Controller is still in service.

- Enable the automatic tap changing for the transformer *T-1*. For this, open the element dialog, navigate to *Load Flow* → *Controller, Tap Changer 1* and enable the option *Automatic Tap Changing*.
- Configure the tap-changer controller as follows:
    - *Tap changer*: discrete
    - *Controlled Node is at*: LV
    - *Control Mode*: V
    - *Voltage Setpoint*: 1.03 p.u.
    - *Lower bound*: 1.02 p.u.
    - *Upper bound*: 1.04 p.u.
- Execute the load flow calculation with the additional option *Automatic tap adjustment of transformers* enabled.
- Are there any warning messages, elements loaded over 100%, or voltage deviations exceeding ± 0.05 p.u.?

Place your answer here →

There is a warning message. No elements loaded over 100%. Both short busbars are exceeding ±0.05 p.u. voltage deviations. Screenshots of the relevant data tables and diagram are as follows.

```
110 kV grid\Station Control.ElmStactrl:
 '— BB1' : Voltage setpoint cannot be kept ( v = 1.0129 p.u., vset = 1.0200 p.u. )
 ---------------------------------------------------------------------------------
```

Figure: Warning

Voltage and Loading Violations

Loading Violations

Maximum Loading of Branch Elements 80 %						
Name	**Element type**	**Loading %**	**Station\Busbar**	**Total Apparent Power MVA**	**Positive-Sequence Current, Magnitude A**	**Positive-Sequence Current, Magnitude p.u.**
T -1	2-Winding Transformer	83.4	380kV Busbar	127.6	190.004	0.834
			SS2\BB2	116.5	602.236	0.765

Figure: Voltage & Loading Violations

- Display the transformer tap positions on the network graphic (use *Layers* and activate the corresponding setting within the *Element Details* page). What is the tap position?

Place your answer here →

Tap position is 1.

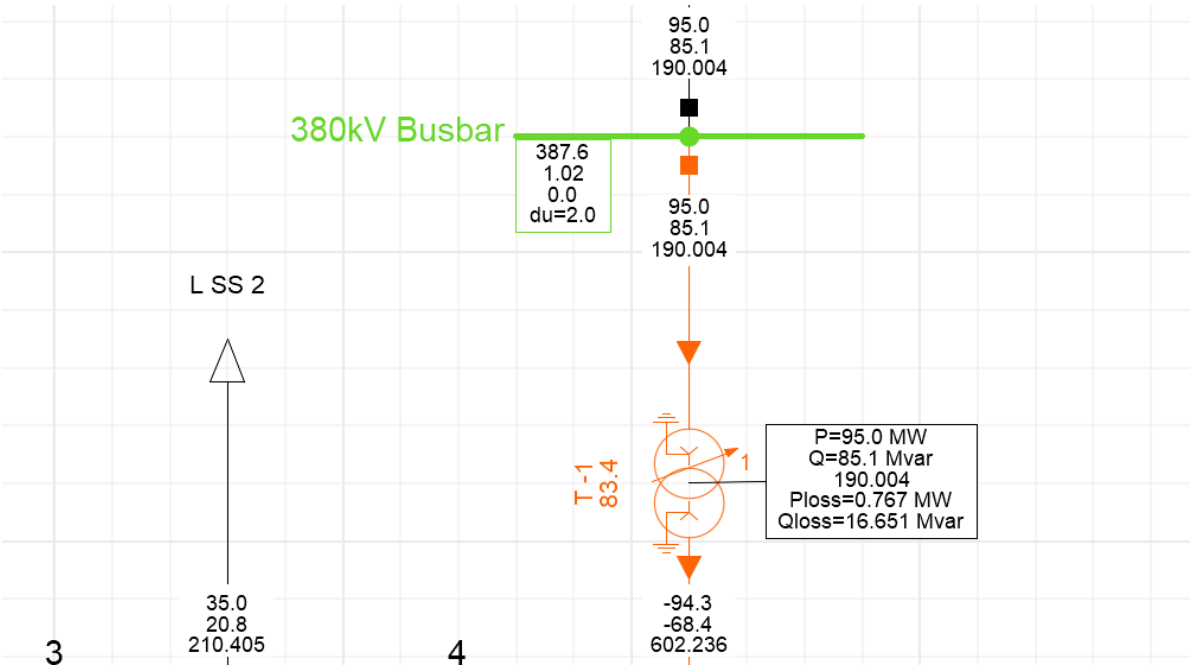


Figure: Tap Position

Task D. Network Enhancement

Within this task, you will firstly become familiar with a 20 kV grid. Afterwards a new line will be entered, then loads and a wind farm will be connected along this line.

- Import and activate the project *Task D*.
- What date does the *Study Case* have? Note that you can change settings using the *Project Overview* window. Change the date of the *Study Case* to the current day.
- Activate the graphic mode to make graphical changes and add the components of the 20 kV grid as shown in Figure 3.

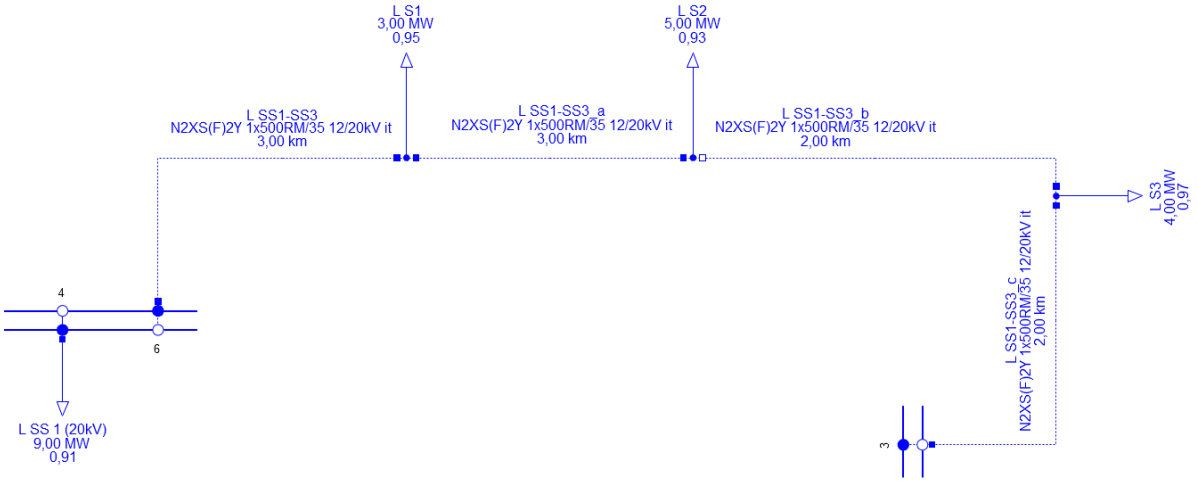


Figure 3: Enhancement of the 20 kV grid

- Add a new line in between *SS1 (20 kV)* and *SS3 (20 kV)*. Connect the line to busbar *S1* of substation *SS1 (20 kV)* (close the switch!) and to busbar *S2.1* of substation *SS3 (20 kV)*. The line has a length of 10 km. Use the line type *N2XS(F)2Y 1x500RM/35 12/20kV* it from the local project library.
- The loads can be connected by selecting the load element and directly clicking on the line. A new window will appear, where you can enter the position of connection. Select the option *Insert Switches* for the *Right Side* and *Left Side* to create switches at the new terminals. To add switches subsequently, right-click on the line very close to the terminal and select *Create Switch*. Enter the load data according to Figure 3.
- Open the circuit breaker on the right side of load L S2 by double-clicking on it.
- Execute a load flow calculation to verify the input data.

- Connect a small wind park, which consists of a wind turbine and a step-up transformer, as shown in Figure 4. Use types from the Project Library for the modelling.

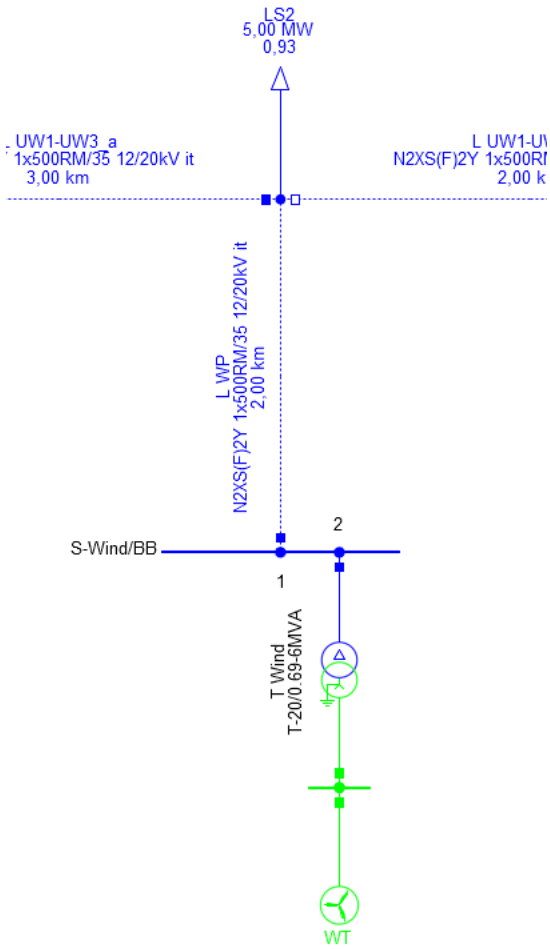


Figure 4: Connection of a wind farm to the medium-voltage grid

- Use a *Single Busbar* system to represent the medium-voltage connection of the wind park:
    - *Name*: S-Wind
    - *Type*: T-SS-20
    - *Nominal Voltage*: 20 kV
- Use a *Busbar (Short)* for the low-voltage side of the wind farm:
    - *Name*: Wind LV
    - *Nominal Voltage*: 0.69 kV
- Connect the *S-Wind* busbar with a *Line*:
    - *Name*: L Wind
    - *Type*: N2XS(F)2Y 1x500RM/35 12/20kV it
    - *Length of Line*: 2.0 km

- Connect the substation *S-Wind* and the busbar *Wind LV* with a *2-Winding Transformer*:
    - *Name*: T Wind
    - *Type*: T-20/0.69kV-6MVA
    - *Number of parallel Transformers*: 2
    - *Unit Transformer* on the *Short-Circuit VDE/IEC* page
- Connect a *Wind Generator* to the busbar *Wind LV*:
    - *Name*: WT
    - *Plant Category*: Wind
    - *Subcategory*: Onshore
    - *Number of parallel units*: 1
    - *Rated Apparent Power*: 5 MVA
    - *Rated Power Factor*: 0.9
    - *Local Controller*: Const. Q
    - *Dispatch Input Mode*: P, cos(phi)
    - *Dispatch Active Power*: 4.5 MW
    - *Dispatch Power Factor*: 1
    - *Power station unit type*: Full size converter
    - *Three-phase faults, Ik"3PF*: 0.144 kA
    - *Steady-state short-circuit Maximum* and *Minimum current contribution*: 0.144 kA
- Execute a load flow calculation. Use the *Heatmap* button in the toolbar to answer whether the design of the equipment in the network is adequate with regard to their loading.

Place your answer here →

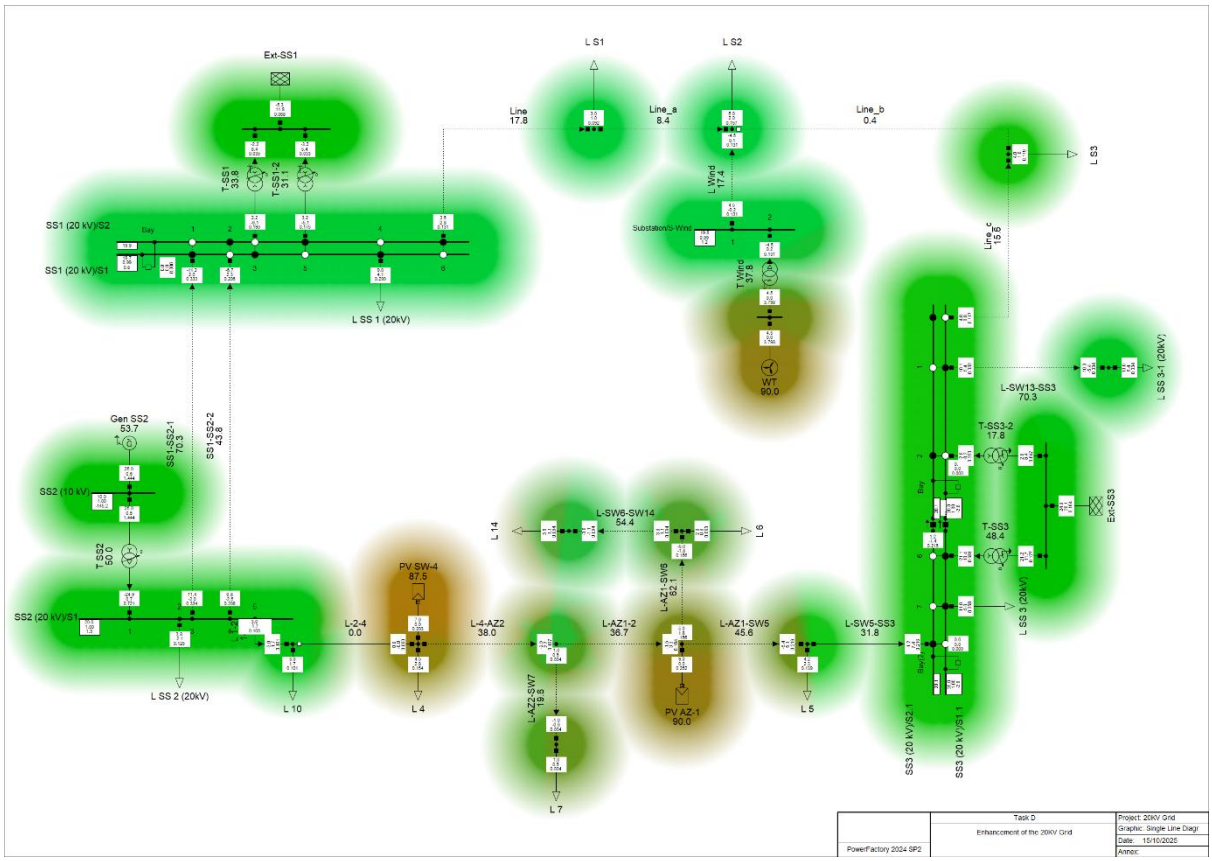


Figure: Heatmap

- It is often the case that a combination of generator, busbar and transformer should be used more than once to represent a plant. You will learn how to create a plant template as a solution to this requirement. Select the wind park created (wind generator, transformer and low-voltage busbar). Right-click on the selection and select *Define Template…* Name the template *Wind Park*.
- Now you will connect the newly created template to the substation *S-Wind*. Press the *General detailed templates* button in the drawing toolbox. A list of available templates will be shown. Select the one called *Wind Park*. Click once into the drawing area. You can see the shape of the selected template attached to your mouse pointer. Place the template with another mouse click to connect another wind park to substation *S-Wind*.

Task E. Connection of Grids

- Import and activate the project *Task E*.
- First look at the Project Overview window: how many networks are active and therefore considered for calculations?

Place your answer here →

There are two active grids which are 110 kV and 20 kV.

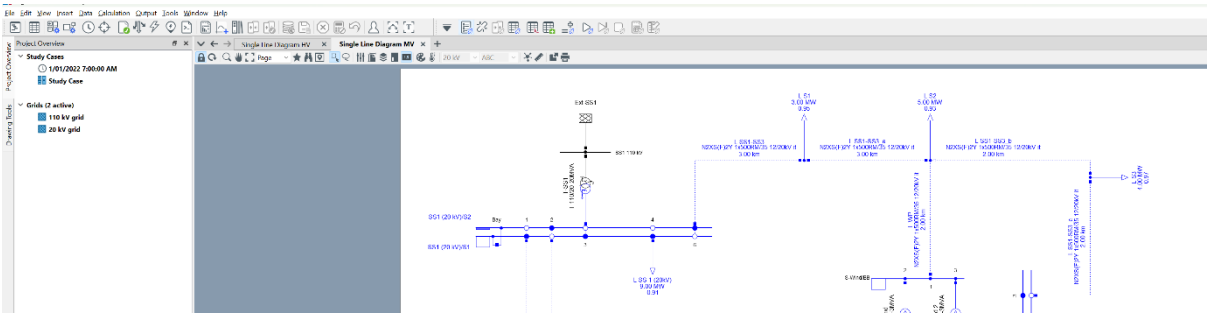


Figure: Connection of Active Grids

- The project contains a 110 kV and 20 kV grid, shown in the diagrams *Single Line Diagram HV* and *Single Line Diagram MV* respectively. The two grids are not connected, but if you look in the Network Model Manager you can see that the 20 kV grid contains two 110 kV busbars. The busbars *SS1 110 kV* and *SS3 110 kV* are currently just connected to external grids but could in fact correspond to *SS1/S1* and *SS3/S1* in the 110 kV grid and be represented on both graphics. Select them in the *Single Line Diagram MV*. You will now use two different methods for achieving the connection just described.
- Delete the external network *Ext-SS1* and the busbar *SS1 110 kV* in the 20 kV grid. Navigate to the 110 kV grid and copy the busbar *SS1/S1*. Return to the 20 kV grid and select the right-click option *Paste Graphically Only*. A graphical representation of the element will be added to the 20 kV grid; the electrical element will remain in the 110 kV grid. Connect the transformer *T-SS1* to the graphically copied busbar by right-clicking on the transformer and selecting the option *Connect...* Validate the results by performing a Load Flow Calculation. The voltage at the busbar *SS1/S1* should be the same for both grids. Right click on the busbar *SS1/S1* and select *Mark in Other Graphic* to jump from one grid to the other.
- Start the second method by deleting the elements *Ext-SS3* and *SS3 110 kV*. Right-click on the transformer *T-SS3* and select the option *Connect...* You can see now a drawing line linked to your mouse. Click with the left mouse in the single line diagram tab *Single Line Diagram HV* to switch the single line diagram. Connect the transformer to the busbar *SS3/S1* in the 110 kV grid.

- Perform a Load Flow Calculation and evaluate the results. Are any elements overloaded?

Place your answer here →

Single Line Diagram MV: PV SW-4 (87.5%), PV AZ-1 (90.0%), WT (90.0%) & WT- 2 (90.0%)

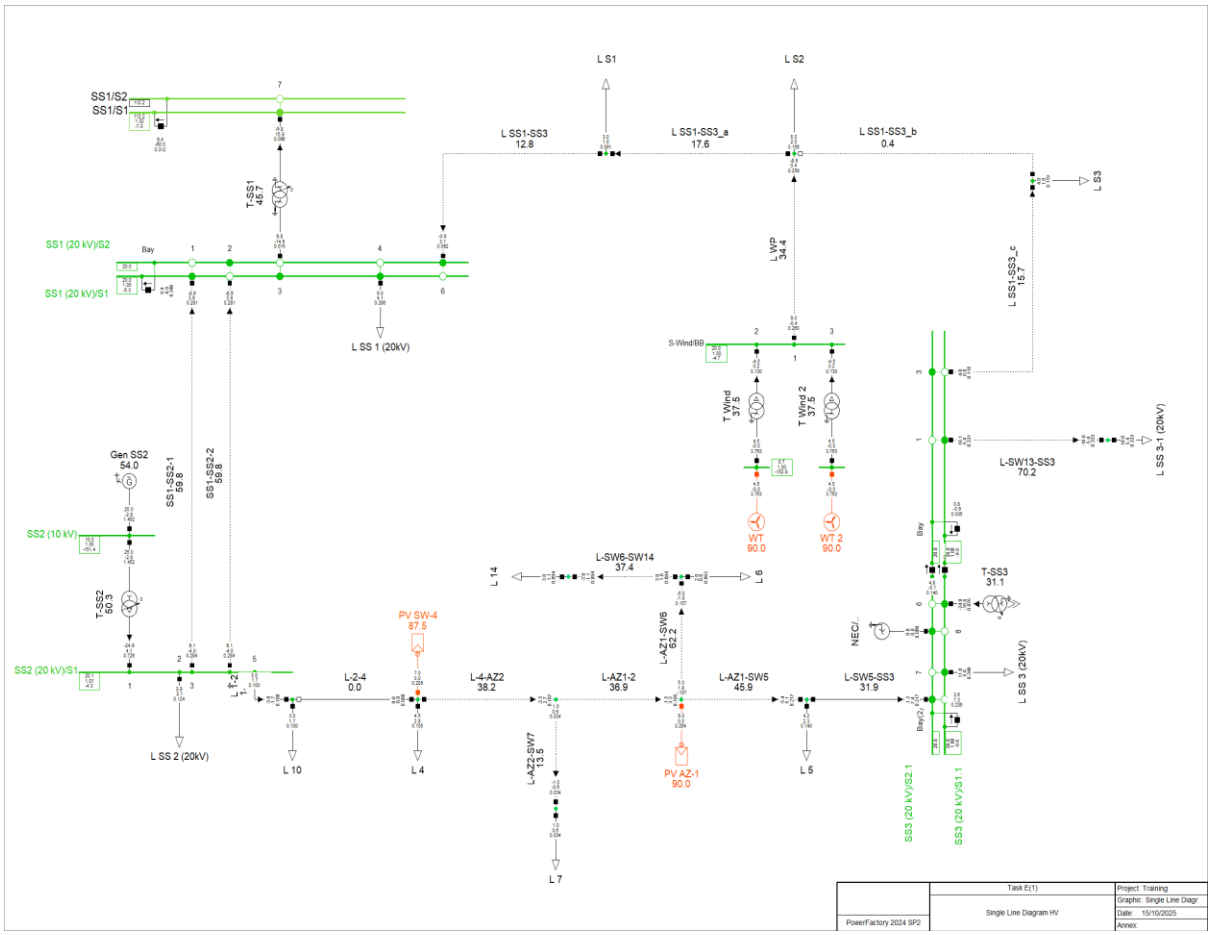


Figure: Single Line Diagram of the 20 kV Grid

There are some overloaded elements in both HV and MV single line diagrams. Single Line Diagram HV: L 2-1 (80.4%), Gen-1 (150.3%), Gen-2 (150.3%), T PP2 (102.7%) and TPP1 (102.7%)

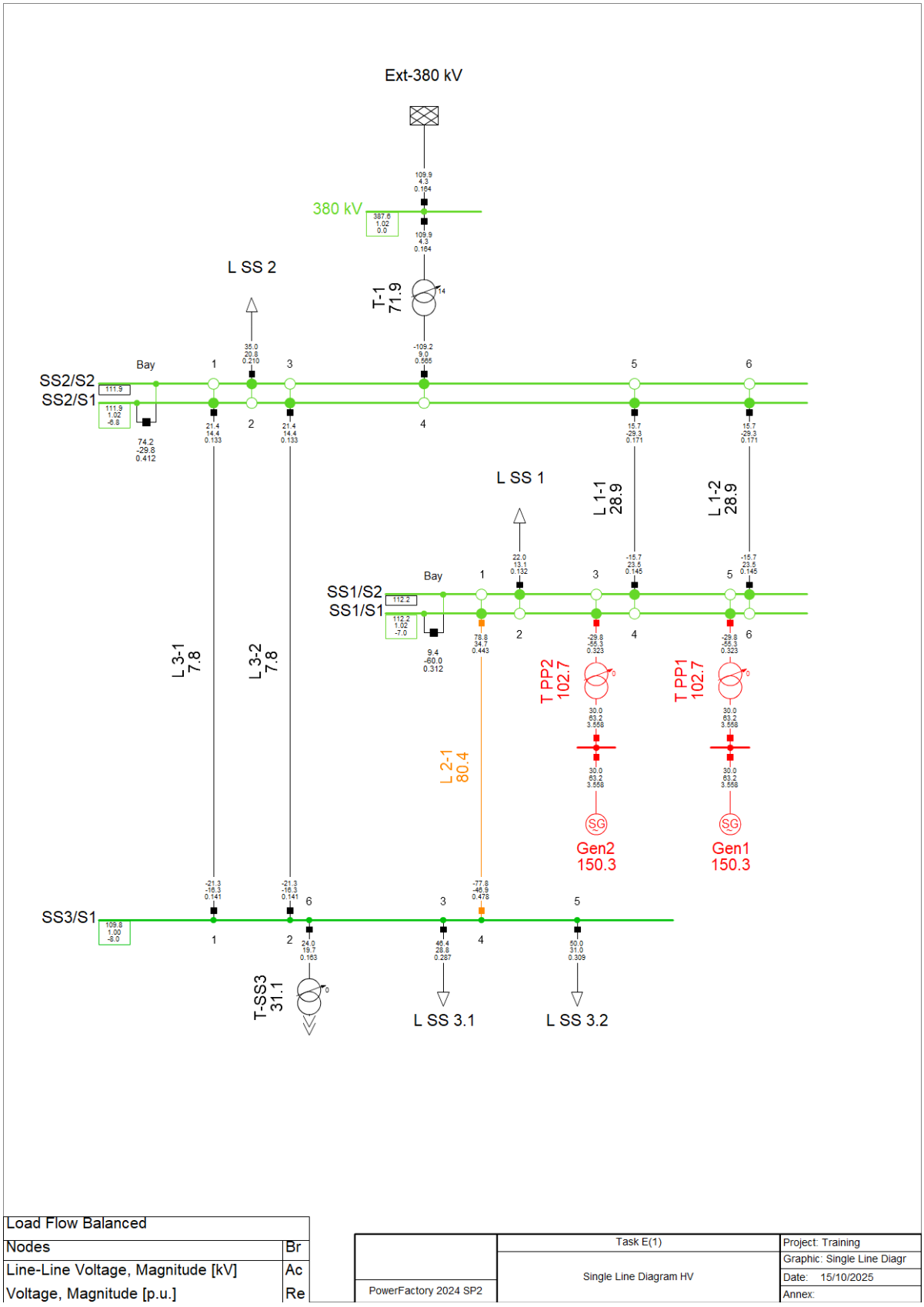


Figure: Single Line Diagram

- What is the interchange power between the two grids? Use the *Network Model Manager* and under *Grid* add the calculation parameters *Total Area Exchange Power*, *Export, Active Power*, *Export, Reactive Power*, *Import, Active Power*, *Import, Reactive Power*.

Place your answer here →

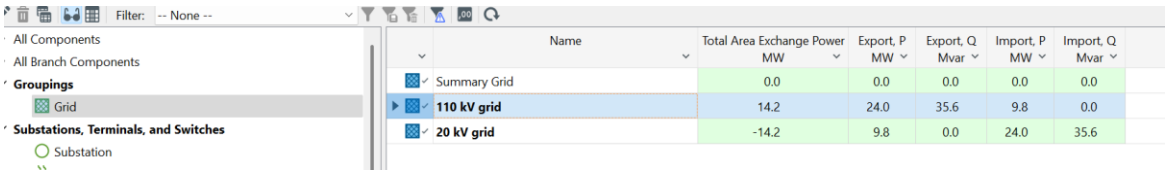


Figure: Total Area Exchange Power Export Active & Reactive Power, Import Active & Reactive Power.

- In addition to the existing schematic diagrams, additional graphics can be generated to represent the network. Open the *Diagram Layout Tool* from the toolbar. By using it new diagrams can be created automatically. To display the interchange power between networks, create a new diagram for the *area interchange type Grids*.
- The individual elements and substations can be displayed not only in the schematic diagram but also in the geographical diagram. The geographical information (location) is stored in node elements (e.g., busbars) and grouping objects (e.g., switchgear). In the following, the substations of the 110 kV network will be displayed in the geographical map. Define the location of the substations in the 110 kV network according to Table 12 by entering the *Geographical Coordinates* on the *Description* page.

Table 12: Geographical position of the substations

Substation	*Latitude / Northing*	*Longitude / Easting*
SS1	-31.8 deg.	115.8 deg.
SS2	-32 deg.	116 deg.
SS3	-32.5 deg.	115.75 deg.

- Create the geographical map via *Insert* → *Geographic Diagram...* It is possible to work simultaneously on the geographical map and in schematic plans. Changes of the graphical information (coordinates, line lengths) can be adopted automatically. Grids/elements created in the geographical map can be inserted/newly created in

schematic plans using the *Diagram Layout Tool*. When lines are drawn in the geographical map, the line length and coordinates will be automatically updated. Existing lines in the geographical map can be measured via right-click → *Measure Length of Lines…* when the *Graphical Freeze Mode* is not active.

Answer:

The total length of the marked lines(s) was 167.702 Km.

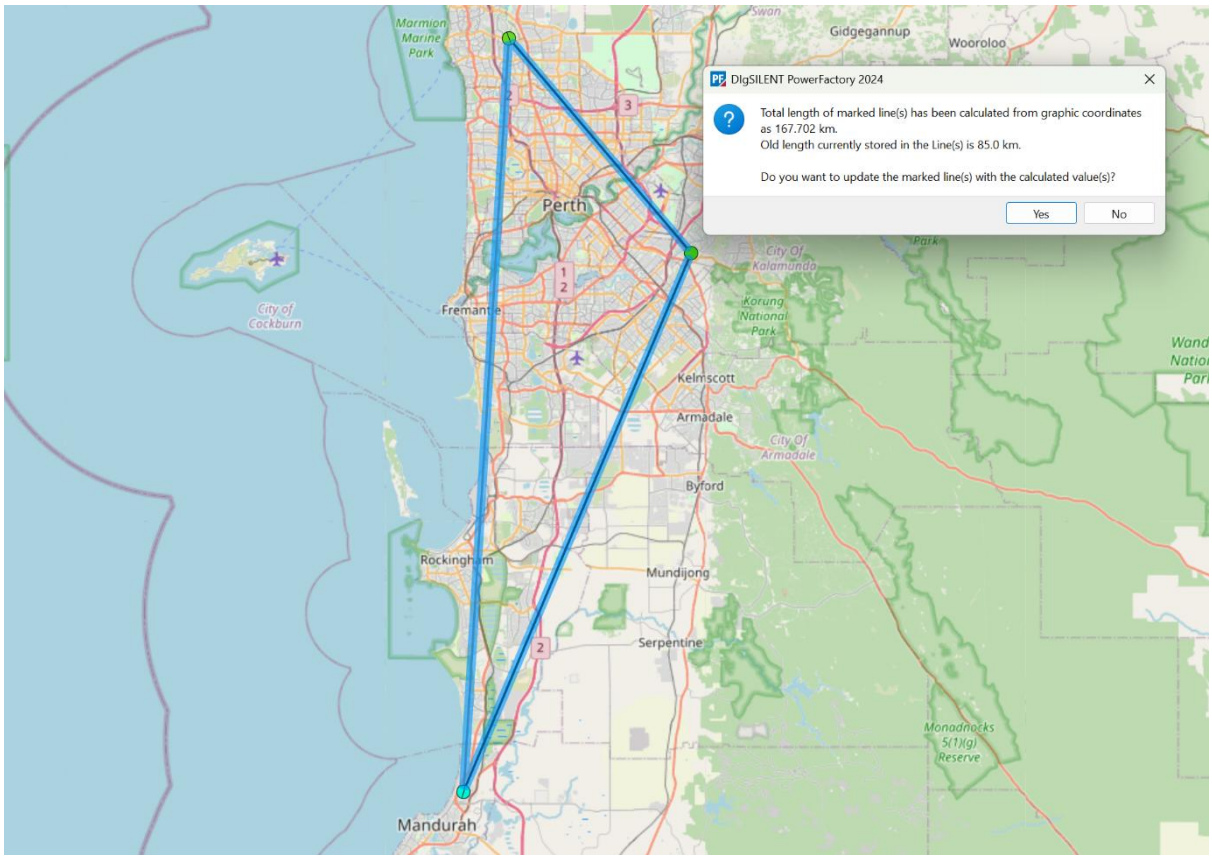
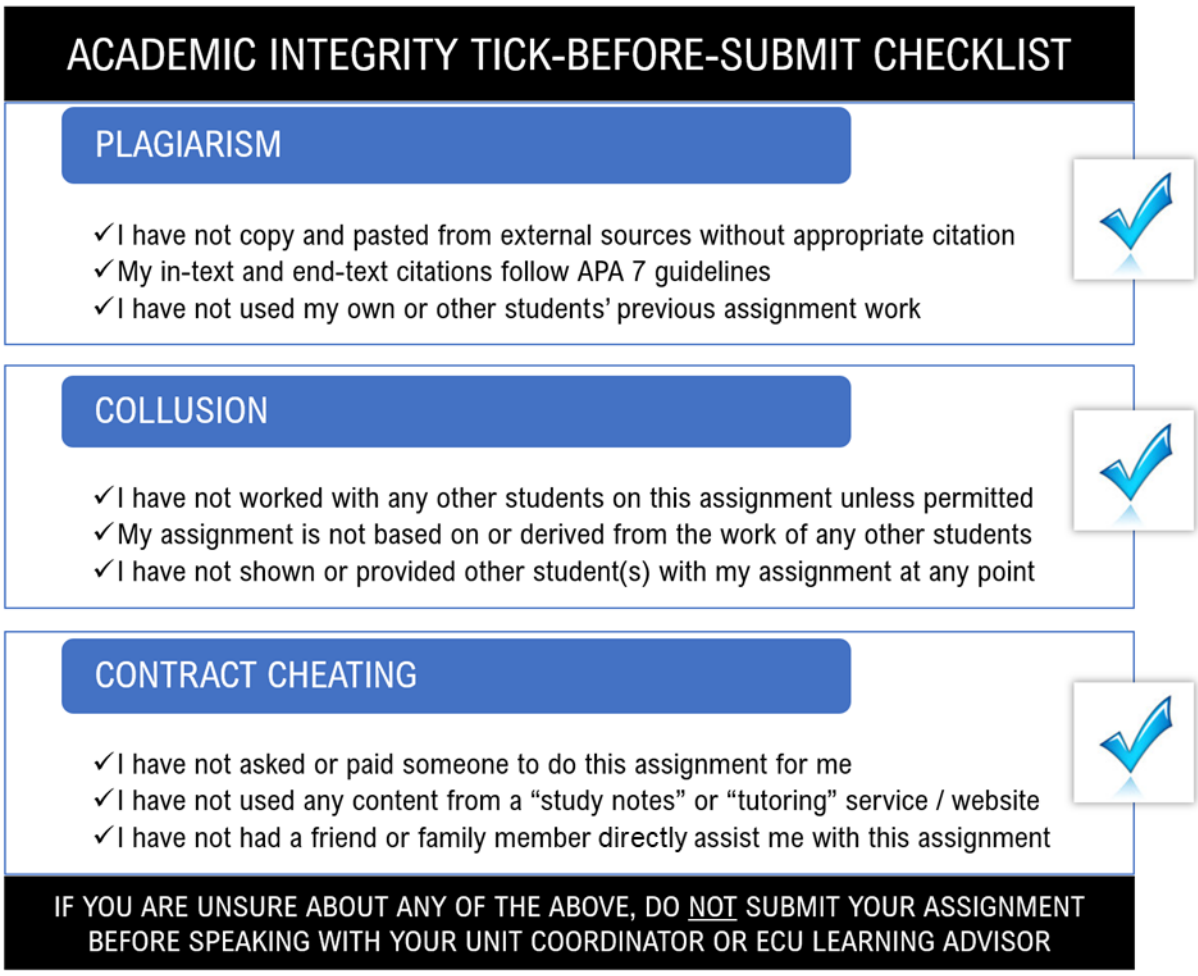


Figure: Length of Lines

N.B.:

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